

Sprint 03: System Integration and Final Delivery Planning

CP317 - A - Group 6

Laurier Club Room Booking System

Team Members:

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Sprint Goal

The goal of Sprint 03 was to advance HawkSpace into a more feature-complete and student-ready platform by adding analytical tools, improving usability, and introducing accessibility improvements. This sprint focused on delivering the Dashboard (timetable + upcoming event widget), implementing recurring bookings, adding override-booking capabilities for administrators, generating room utilization reports, and meeting WCAG accessibility requirements.

These features move HawkSpace closer to a realistic club-management system by enabling better scheduling visibility, inclusive design, and more flexible administrative operations. Sprint 03 builds directly on the structure established in earlier sprints, preparing the application for future enhancements such as reminders, deeper analytics, and calendar-based navigation.

Sprint Backlog Summary

During Sprint 03, the team completed all major planned stories. The Dashboard Overview feature was fully implemented, providing users with a clean visual summary of their weekly timetable along with an automatic “Next Upcoming Event” highlight. The Override Booking feature was completed, allowing administrators to replace conflicting bookings while maintaining a detailed audit log.

Recurring booking functionality was also finished, enabling club executives to schedule weekly or monthly meetings without repeatedly entering details. The Room Utilization Report was implemented with aggregated booking data and a finalized chart/table visualization. Lastly, the Accessibility Enhancements story was fully completed—screen-reader labels, semantic structure, and high-contrast mode were all added to meet WCAG 2.1 AA guidelines.

All stories assigned to Sprint 03 reached completion, with team members contributing across frontend design, backend logic, and system integration.

Implementation Progress

The Dashboard feature was successfully designed and implemented, including timetable rendering, data synchronization, and an upcoming-event selector that automatically identifies the next scheduled booking for the logged-in user.

The Recurring Booking system is now fully functional. Users can select weekly or monthly recurrence patterns, and the system automatically generates the full event series while performing conflict checks. Editing and cancellation of recurring series were implemented with consistent behavior across the UI.

The Room Utilization Report aggregates usage trends by day and time. A responsive chart + table view was added that allows administrators to quickly identify high-demand rooms and peak booking periods.

The Override Booking workflow was completed with a full admin UI, confirmation modal, and audit logging. All override actions write structured event entries to the database for transparency.

The Accessibility Enhancements are fully integrated into the platform. ARIA labels, improved semantic structure, keyboard navigation fixes, and a toggleable high-contrast mode together ensure that the platform is inclusive for users with visual and mobility impairments.

UX improvements from previous sprints were expanded upon, adding padding fixes, layout consistency, and improved responsive scaling for mobile devices.

Technical Challenges and Resolutions

The team encountered several challenges. Handling recurring bookings required designing automatic conflict detection to prevent overlapping events. This was resolved by adding a utility function that validates the entire event series before creation.

Real-time dashboard updates initially caused stale displays, but this was fixed by integrating snapshot listeners that refresh data when the database changes.

Rendering the room utilization chart introduced performance issues on lower-power devices. The team tested multiple options and settled on a lightweight charting library that balanced clarity and performance.

Accessibility testing revealed several ARIA and contrast issues, which were resolved by adjusting label structures, refining heading hierarchy, and implementing a dark/high-contrast theme variant.

Testing and Review

Manual testing was performed across all completed features. Dashboard rendering, upcoming-event logic, and timetable accuracy were validated using multiple test accounts. Recurring booking behavior was tested with weekly and monthly patterns, including editing and deleting series.

Override functionality was tested to ensure the replaced event is properly removed and logged.

The Room Utilization Report was tested with a range of sample booking data to validate both chart and table accuracy.

Accessibility testing was performed using NVDA and browser-based Lighthouse audits, confirming compliance with WCAG 2.1 AA for contrast, labels, and keyboard navigation.

Overall, all test cases passed, and the behavior matched expected outcomes for each story.

Product Owner Feedback

The Product Owner expressed satisfaction with the dashboard layout and the improved visibility it offers to end users. Recurring booking functionality was identified as a major quality-of-life improvement for club executives. The room

utilization report meets the administrative need for analytics, and the override functionality adds flexibility for managing priority events.

Accessibility improvements were praised as essential for broad usability and better alignment with campus accessibility expectations. Some minor polishing for responsiveness and visual consistency was noted for future refinement.

Next Steps (Sprint 04 final)

The next sprint will focus on integrating advanced analytics, Additional automated tests will be added and performance enhancements will be prioritized to ensure scalability as booking volume grows.